START Program

OUTPUT: 'WELCOME TO HOSPITAL MANAGEMENT SYSTEM'

Allocate dynamic memory for patients array

Is patients memory allocated?

No

OUTPUT: 'Memory allocation failed!'

END WITH CODE '

Yes

Call loadHospitalData

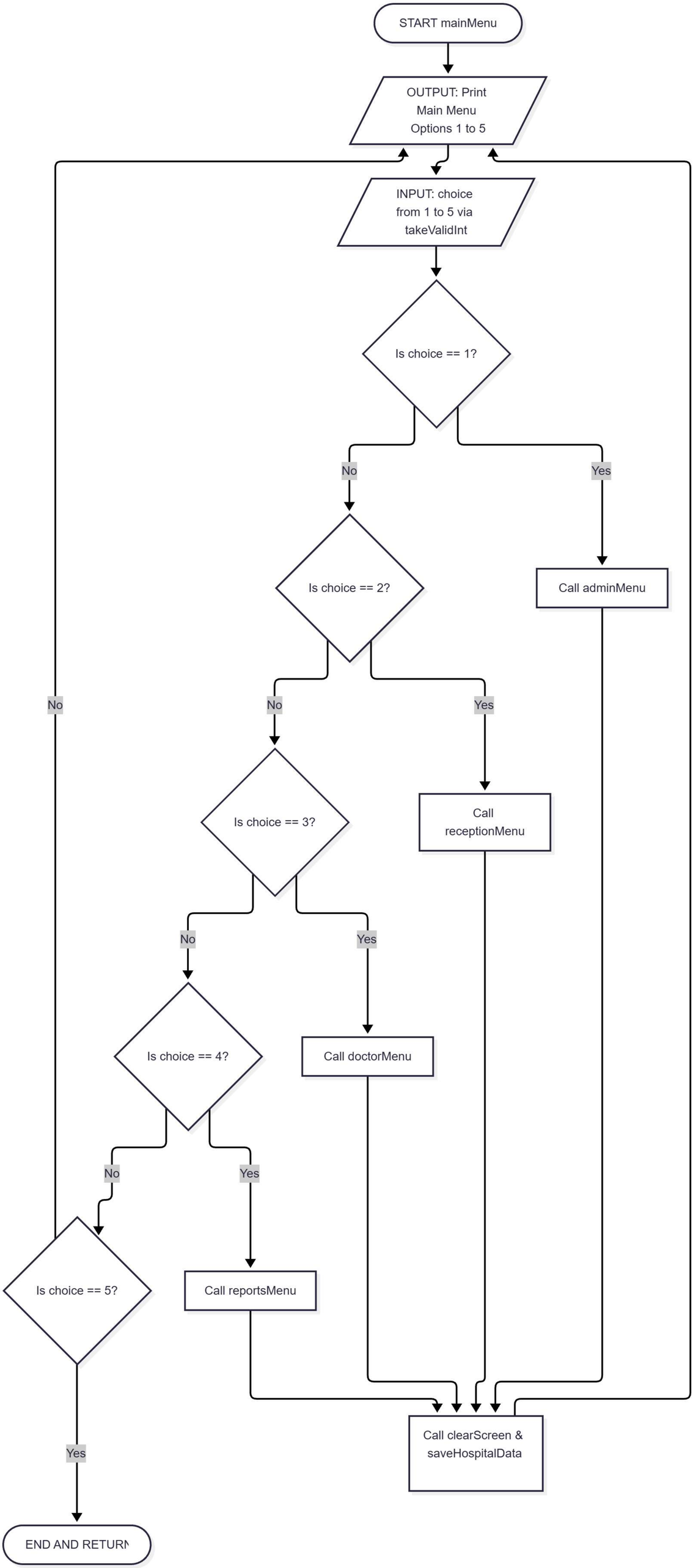
Call mainMenu

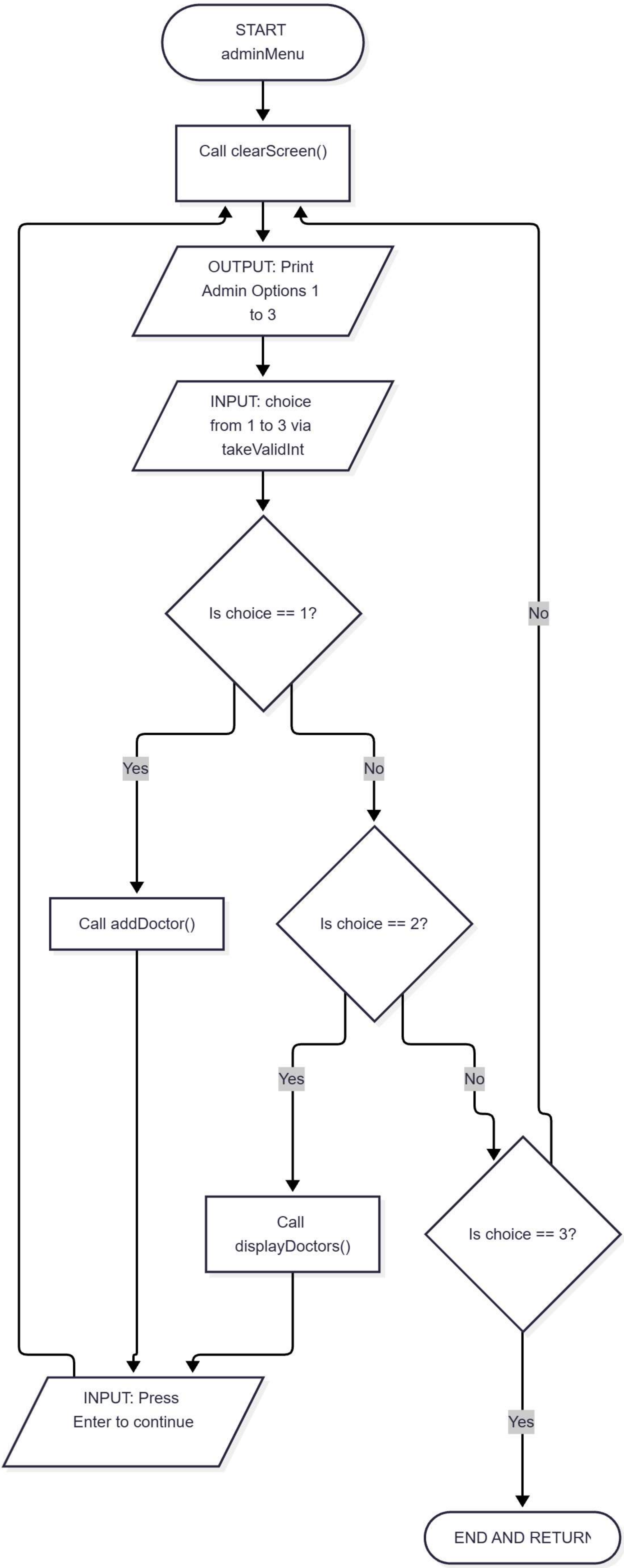
Save Hospital Data to File

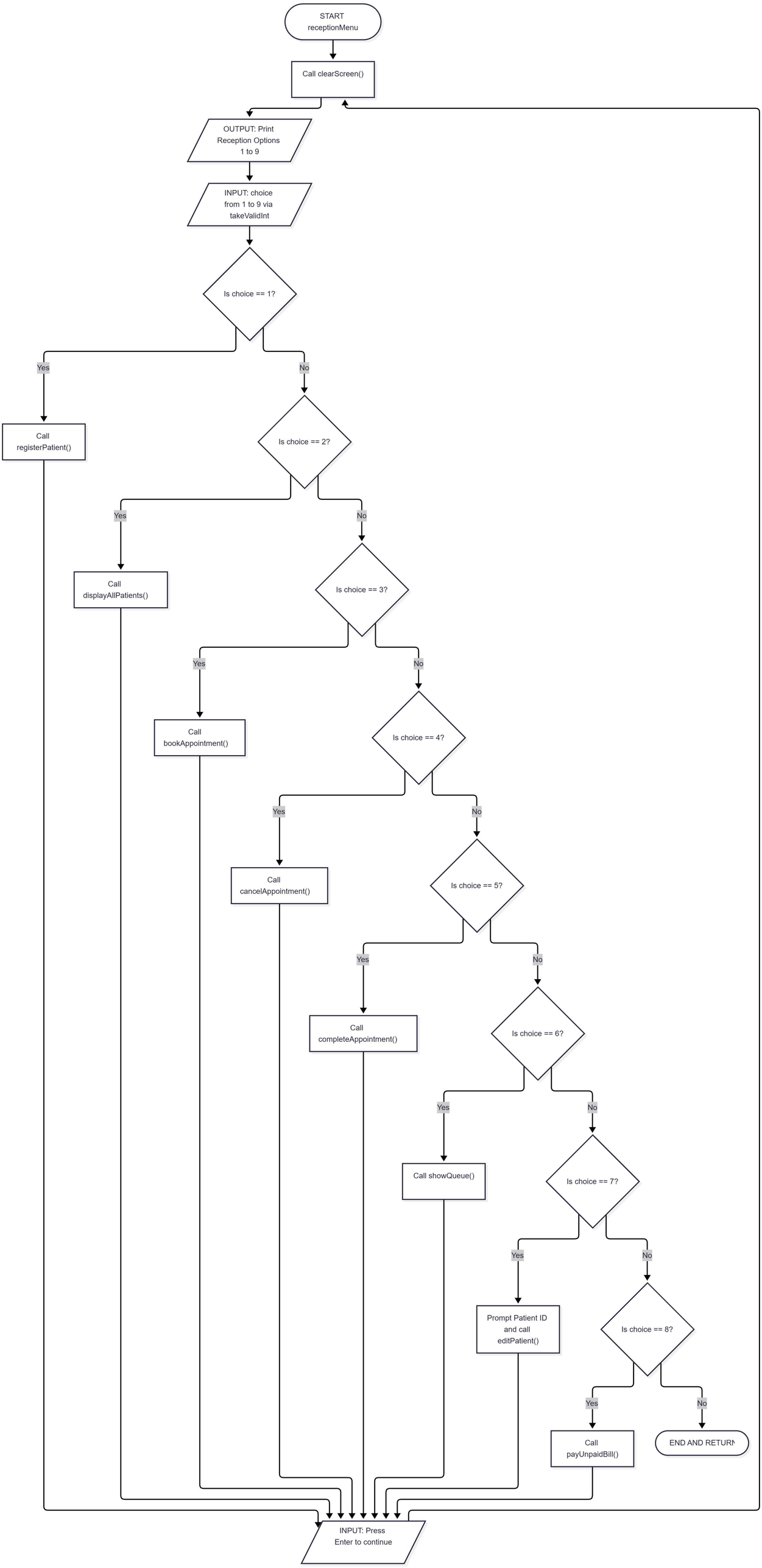
Free dynamically allocated patients memory

OUTPUT: 'Thank you for using system. Exiting...'

END WITH CODE (


START mainMenu
OUTPUT: Print Main Menu Options 1 to 5
INPUT: choice from 1 to 5 via takeValidInt
Is choice == 1?
Yes
Call adminMenu
No
Is choice == 2?
Yes
Call receptionMenu
No
Is choice == 3?
Yes
Call doctorMenu
No
Is choice == 4?
Yes
Call reportsMenu
No
Is choice == 5?
Yes
END AND RETURN
Call clearScreen & saveHospitalData


START
adminMenu
Call clearScreen()
OUTPUT: Print
Admin Options 1
to 3
INPUT: choice
from 1 to 3 via
takeValidInt
Is choice == 1?
No
Yes
No
Call addDoctor()
Is choice == 2?
Yes
No
Call
displayDoctors()
Is choice == 3?
INPUT: Press
Enter to continue
Yes
END AND RETURN


START
receptionMenu
Call clearScreen()
OUTPUT: Print
Reception Options
1 to 9
INPUT: choice
from 1 to 9 via
takeValidInt
Is choice == 1?
Yes
No
Call
registerPatient()
Is choice == 2?
Yes
No
Call
displayAllPatients()
Is choice == 3?
Yes
No
Call
bookAppointment()
Is choice == 4?
Yes
No
Call
cancelAppointment()
Is choice == 5?
Yes
No
Call
completeAppointment()
Is choice == 6?
Yes
No
Call showQueue()
Is choice == 7?
Yes
No
Prompt Patient ID
and call
editPatient()
Is choice == 8?
Yes
No
Call
payUnpaidBill()
END AND RETURN
INPUT: Press
Enter to continue

START
doctorMenu

Call clearScreen()

OUTPUT: Print
Doctor Options 1
to 2

INPUT: choice
from 1 to 2 via
takeValidInt

Is choice == 1?

Yes

No

Call
showSchedule()

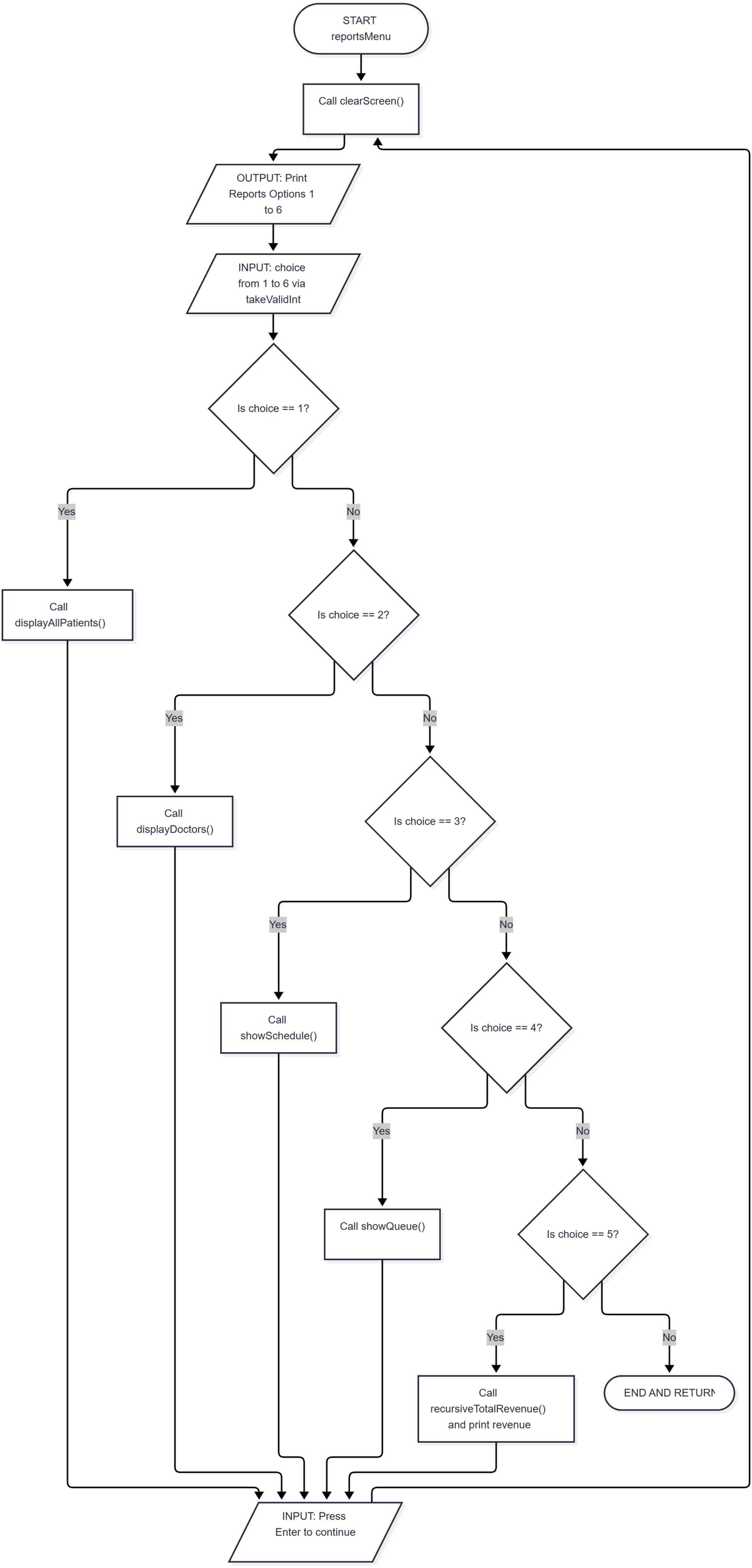
Is choice == 2?

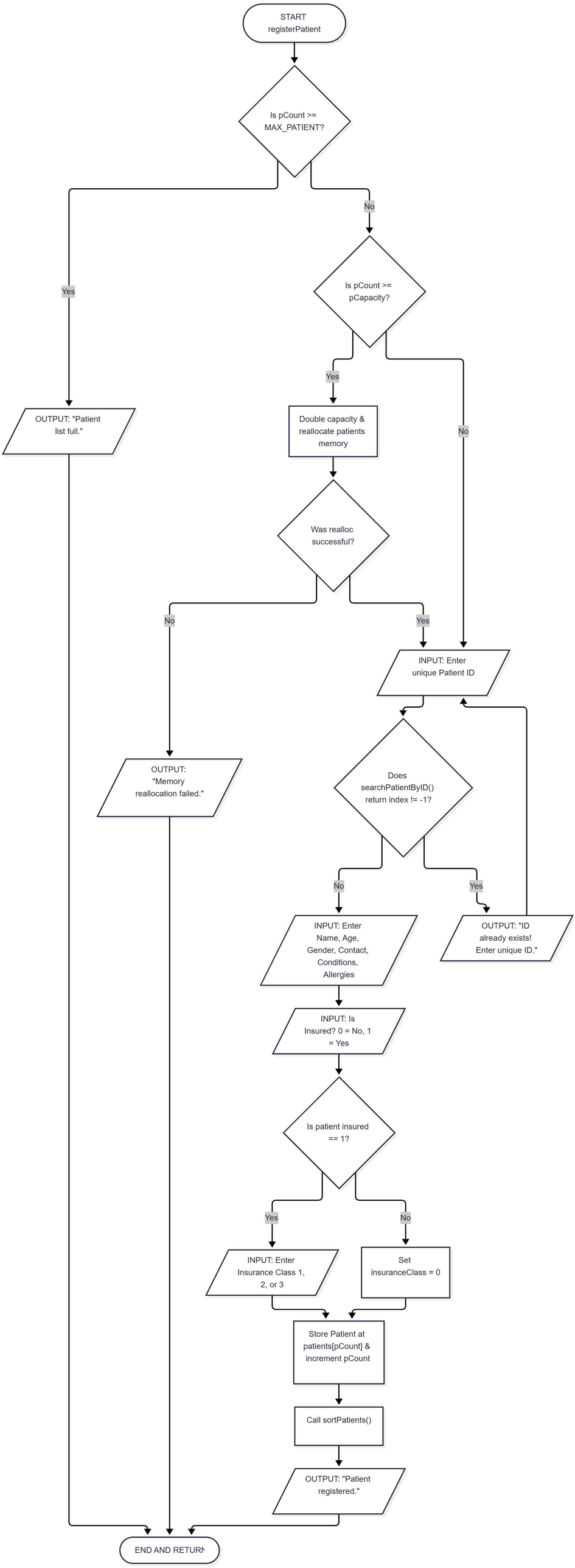
No

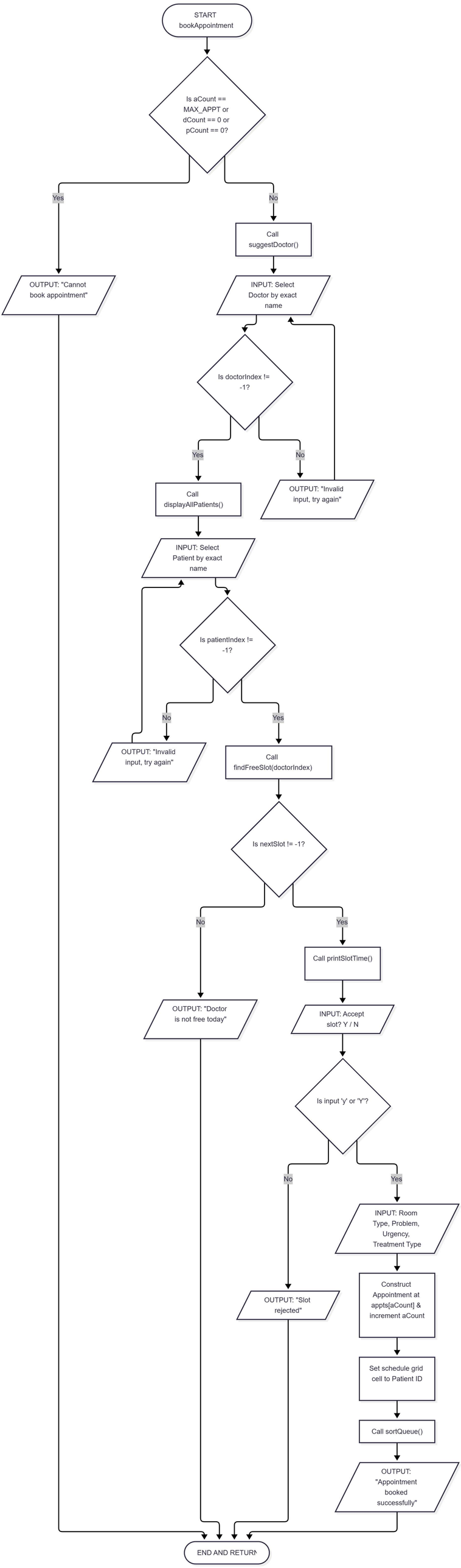
Yes

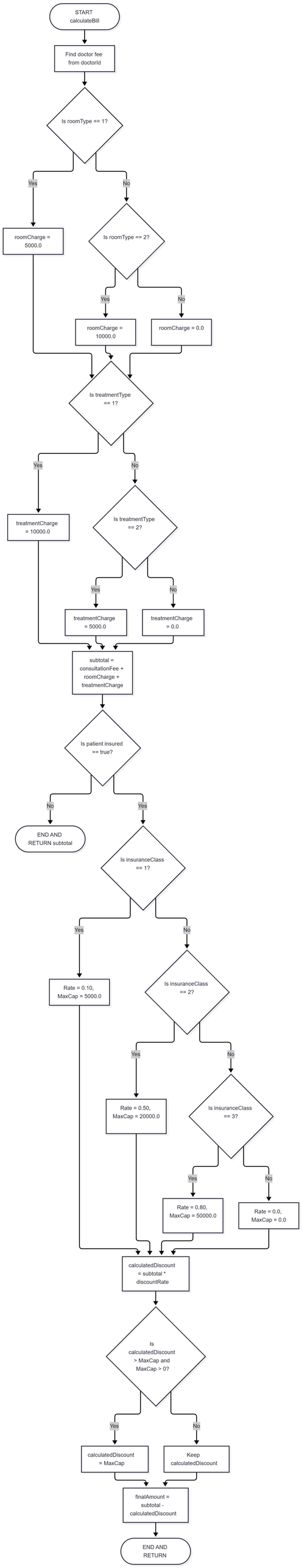
INPUT: Press
Enter to continue

END AND RETURN

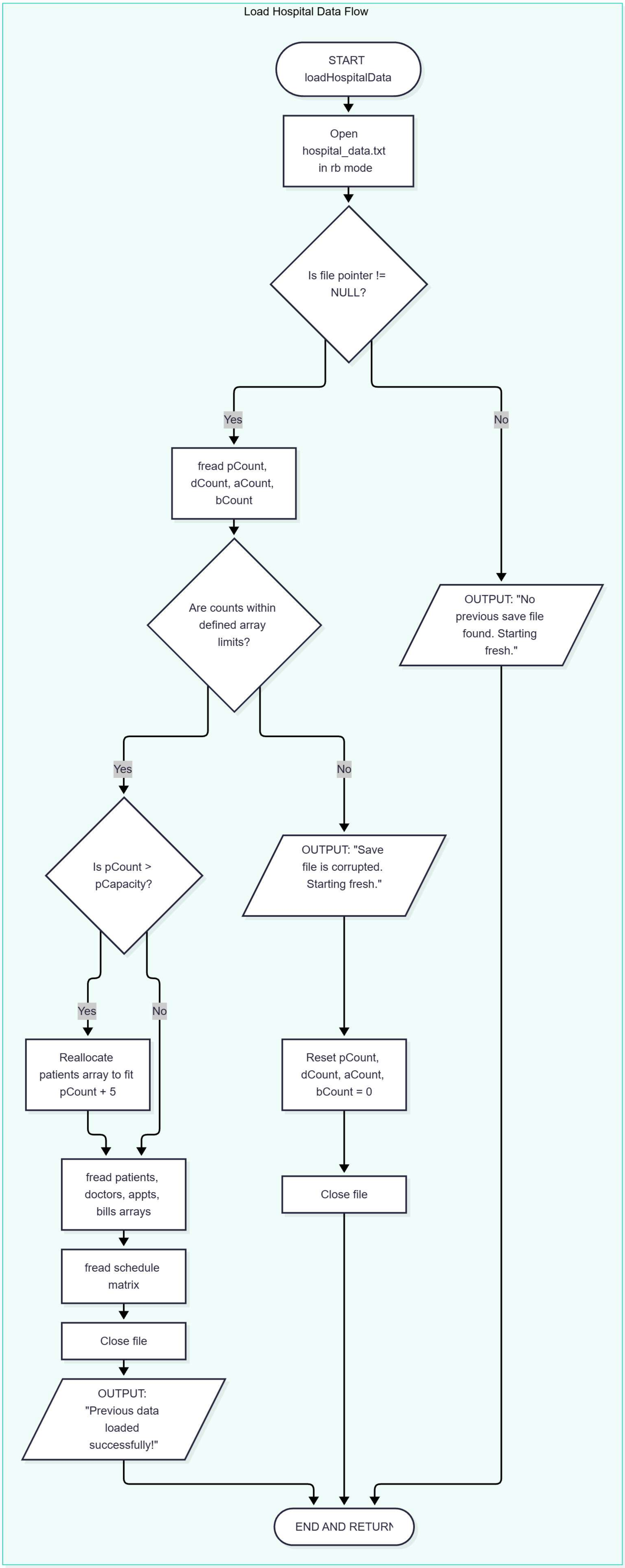

START
reportsMenu
Call clearScreen()
OUTPUT: Print
Reports Options 1
to 6
INPUT: choice
from 1 to 6 via
takeValidInt
Is choice == 1?
Yes
No
Call
displayAllPatients()
Is choice == 2?
Yes
No
Call
displayDoctors()
Is choice == 3?
Yes
No
Call
showSchedule()
Is choice == 4?
Yes
No
Call showQueue()
Is choice == 5?
Yes
No
Call
recursiveTotalRevenue()
and print revenue
END AND RETURN
INPUT: Press
Enter to continue


START registerPatient
Is pCount >= MAX_PATIENT?
Yes
No
OUTPUT: "Patient list full."
Is pCount >= pCapacity?
Yes
No
Double capacity & reallocate patients memory
Was realloc successful?
No
Yes
OUTPUT: "Memory reallocation failed."
INPUT: Enter unique Patient ID
Does searchPatientByID() return index != -1?
No
Yes
OUTPUT: "ID already exists! Enter unique ID."
INPUT: Enter Name, Age, Gender, Contact, Conditions, Allergies
INPUT: Is Insured? 0 = No, 1 = Yes
Is patient insured == 1?
Yes
No
INPUT: Enter Insurance Class 1, 2, or 3
Set insuranceClass = 0
Store Patient at patients[pCount] & increment pCount
Call sortPatients()
OUTPUT: "Patient registered."
END AND RETURN


START bookAppointment
Is aCount == MAX_APPT or dCount == 0 or pCount == 0?
Yes
No
OUTPUT: "Cannot book appointment"
Call suggestDoctor()
INPUT: Select Doctor by exact name
Is doctorIndex != -1?
Yes
No
Call displayAllPatients()
OUTPUT: "Invalid input, try again"
INPUT: Select Patient by exact name
Is patientIndex != -1?
No
Yes
OUTPUT: "Invalid input, try again"
Call findFreeSlot(doctorIndex)
Is nextSlot != -1?
No
Yes
OUTPUT: "Doctor is not free today"
Call printSlotTime()
INPUT: Accept slot? Y / N
Is input 'y' or 'Y'?
No
Yes
OUTPUT: "Slot rejected"
INPUT: Room Type, Problem, Urgency, Treatment Type
Construct Appointment at appts[aCount] & increment aCount
Set schedule grid cell to Patient ID
Call sortQueue()
OUTPUT: "Appointment booked successfully"
END AND RETURN


START calculateBill
Find doctor fee from doctorId
Is roomType == 1?
Yes
No
roomCharge = 5000.0
Is roomType == 2?
Yes
No
roomCharge = 10000.0
roomCharge = 0.0
Is treatmentType == 1?
Yes
No
treatmentCharge = 10000.0
Is treatmentType == 2?
Yes
No
treatmentCharge = 5000.0
treatmentCharge = 0.0
subtotal = consultationFee + roomCharge + treatmentCharge
Is patient insured == true?
No
Yes
END AND RETURN subtotal
Is insuranceClass == 1?
Yes
No
Rate = 0.10, MaxCap = 5000.0
Is insuranceClass == 2?
Yes
No
Rate = 0.50, MaxCap = 20000.0
Is insuranceClass == 3?
Yes
No
Rate = 0.80, MaxCap = 50000.0
Rate = 0.0, MaxCap = 0.0
calculatedDiscount = subtotal * discountRate
Is calculatedDiscount > MaxCap and MaxCap > 0?
Yes
No
calculatedDiscount = MaxCap
Keep calculatedDiscount
finalAmount = subtotal - calculatedDiscount
END AND RETURN

Load Hospital Data Flow



Save Hospital Data Flow

